Testing features

CodePen offers several testing features that help developers validate and debug their code. These features are designed to assist in the process of identifying and resolving issues or errors in web projects.

- ***Real-time preview:*** CodePen provides a live preview of your project, allowing you to instantly see the results of your code changes. This real-time feedback helps you quickly identify any visual issues or unexpected behaviour as you modify your HTML, CSS, and JavaScript code.
- ***Console output:*** CodePen includes a console panel where you can view and interact with the JavaScript console output. This is useful for debugging purposes, as you can log messages, inspect variable values, and track runtime errors.
- ***Browser compatibility testing:*** CodePen's 'browsers' feature enables you to test your project in different web browsers and versions. This allows you to ensure that your code functions correctly and appears as intended across various browser environments.
- ***Debugging tools:*** CodePen provides debugging tools that assist in diagnosing and resolving issues in your JavaScript code. These tools include breakpoints, step-by-step code execution, and variable inspection.
- ***Error reporting:*** If your code encounters errors, CodePen highlights the affected lines and provides error messages in the editor. This helps you quickly identify syntax errors, missing variables, or other coding mistakes.
- ***External API testing:*** CodePen allows you to make HTTP requests to external APIs directly from your code. This feature enables you to test the integration of your web project with external services and validate the data or functionality provided by the APIs.
- ***Code validation:*** CodePen includes built-in validators for HTML, CSS, and JavaScript. These validators check your code against industry standards and best practices, highlighting any errors or warnings.

Debugging web applications with CodePen

Debugging web applications with CodePen can be a straightforward and efficient process. The following points illustrate how developers can effectively debug the web application using CodePen.

- **Inspecting elements**: CodePen provides an integrated element inspector that allows you to examine the HTML structure and CSS styles of your web application. Right-click on any element in the live preview and select 'inspect' to open the inspector panel. Here, you can view and modify the HTML markup and CSS styles, helping you identify any issues or discrepancies.
- **Using the console**: CodePen includes a console panel where you can view and interact with the JavaScript console output. The console is invaluable for debugging JavaScript code. You can log messages, inspect variable values, track errors, and run *ad-hoc* JavaScript commands. Use *console.log()* statements to output relevant information and verify that your code is executing as expected.
- **Setting breakpoints**: Breakpoints are markers that pause the execution of your JavaScript code at specific lines. CodePen supports breakpoints in JavaScript code, allowing you to halt execution at crucial points and inspect the state of your variables and objects. Simply click on the line number in the JavaScript editor to set a breakpoint, and the code execution will stop at that line when reached.
- **Stepping through code**: Once a breakpoint is reached, you can use the 'Step Into', 'Step Over', and 'Step Out' buttons in the debugger panel to navigate through your code line by line. This allows you to observe the flow of execution and identify any unexpected behaviour or errors.
- **Inspecting variables**: While debugging, you can inspect the values of variables and expressions. The debugger panel in CodePen provides a list of variables in the current scope, allowing you to view their values and track how they change during code execution. This helps you understand the state of your application and identify any issues related to variable assignments or calculations.
- **Debugging network requests**: If your web application makes API requests or fetches external resources, CodePen's network panel allows you to monitor and debug these requests. You can inspect the request and response headers, check for errors, and analyse the data being transmitted. This helps in identifying issues related to network connectivity, incorrect API calls, or data parsing. END

References

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